



















# Resonant absorption as a direction filter

- ★ If, however,  $B^*$  is already moving, the photon energy in the laboratory frame depends on its emission direction
- ★ A photon emitted along the motion of  $B^*$  is Doppler shifted upward and may acquire exactly the energy required for

$$B + \gamma \rightarrow B^*$$

- ★ A photon emitted opposite to the motion of  $B^*$  is shifted downward and can not be absorbed resonantly

Resonant absorption selects photons emitted along the recoil direction of  $B^*$ .

**Resonant absorption correlates  
photon and neutrino directions**